

Practical 5

Name:Pragati Santosh Gaikwad

PRN:202501040136

The screenshot shows the CODETANTRA course interface. The top navigation bar includes the logo, home link, and user profile. The main header is 'Essentials of Data Science Laboratory - 2304102L - 2026'. The left sidebar lists course units, with 'Practical 5' selected. The main content area is titled 'Practical 5' and contains two sections: 'Practice Lab Assignment' and 'Lab Assignment', both marked as 'Unit - 100% completed'.

The screenshot shows the lab assignment interface for '5.1.1. Stacked Plot'. The task description asks the user to create a stacked area plot for temperature variations across three cities (A, B, and C) over the months of the year. The interface includes a code editor with a Python script using pandas and matplotlib. The script defines a dictionary of monthly temperatures for three cities and uses the 'melt' function to reshape the data. The 'stackplot' function is used to create the stacked area plot. The output shows a 'Test case 1' passed, with a 'Expected output' and 'Actual output' section displaying the resulting stacked area plot.

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Month and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July',
7             'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [15, 18, 22, 25, 28, 30, 32, 35, 30, 25, 20, 18],
9     'City_B_Temperature': [20, 25, 30, 35, 38, 40, 42, 45, 40, 35, 30, 25],
10    'City_C_Temperature': [10, 12, 15, 18, 20, 22, 25, 28, 25, 20, 15, 10]
11 }
12
13 # Write your code...
14 df = pd.DataFrame(data)
15 plt.stackplot(df['Month'], df['City_A_Temperature'],
16             df['City_B_Temperature'], df['City_C_Temperature'])
17 plt.show()
```

Course

mitaoe.codetantra.com/secure/course.jsp?euid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/677cdd52372a227b32ab1c

CodeTantra Teach &... Adobe Acrobat Course

CODETANTRA Home Learn Anywhere

202501040136@mitaoe.ac.in Support Logout

5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:
The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:
To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:
Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as `"Passenger Class Distribution"`.
- Label the x-axis as `"Pclass"` and the y-axis as `"Count"`.

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as `"Gender Distribution"`.

Histogram (Age Distribution):

titanicDat...

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 # Load the titanic dataset from the csv file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 import pandas as pd
11 import matplotlib.pyplot as plt
12
13 # Load the Titanic dataset from the csv file
14 df = pd.read_csv('titanic.csv')
15
16 # Set up the figure for 5 subplots
17 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
```

Average time: 1.624 s, Maximum time: 1.624 s, 1 out of 1 shown test case(s) passed

Test case 1: 1.624 ms, Debug, Expected output, Actual output

Terminal, Test cases

36°C Clear

Search

ENG IN

22:56 03-05-2025

Course

mitaoe.codetantra.com/secure/course.jsp?euid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/677cdd52372a227b32ab1c

CodeTantra Teach &... Adobe Acrobat Course

CODETANTRA Home Learn Anywhere

202501040136@mitaoe.ac.in Support Logout

5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 50 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title 'Age Distribution' to the histogram.

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21371	7.25	0	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17596	71.00	85	C
3	1	3	Hachikian, Miss. Lusine	female	25	0	0	100760	16.00	0	
4	1	1	Heikkinen, Mrs. Teague (Mrs. Lily May Pasi)	female	26	1	0	115063	53.10	132	S
5	0	3	Allison, Mr. Hiram Henry	male	29	0	0	174326	8.65	0	
6	0	3	Harmon, Mr. James (Mr. James)	male	30	0	0	44884	0	0	
7	0	1	McCarty, Mr. Timothy J	male	54	0	0	27062	21.00	0	
8	0	3	Palsson, Anders (Anders)	male	27	1	1	42009	21.00	0	
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	0	94703	11.00	0	
10	1	2	Nasser, Mrs. Nicholas (Adela Gonen)	female	14	1	0	207710	30.00	0	

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, ParCh, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21371,7.25,0,
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17596,71.00,85,C
3,1,3,"Hachikian, Miss. Lusine",female,25,0,0,100760,16.00,0
4,1,1,"Heikkinen, Mrs. Teague (Mrs. Lily May Pasi)",female,26,1,0,115063,53.10,132,S
5,0,3,"Allison, Mr. Hiram Henry",male,29,0,0,174326,8.65,0
6,0,3,"Harmon, Mr. James (Mr. James)",male,30,0,0,44884,0
7,0,1,"McCarty, Mr. Timothy J",male,54,0,0,27062,21.00,0,0
8,0,3,"Palsson, Anders (Anders)",male,27,1,1,42009,21.00,0
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,0,94703,11.00,0
10,1,2,"Nasser, Mrs. Nicholas (Adela Gonen)",female,14,1,0,207710,30.00,0,C
```

Histogram...

```
1 # import pandas as pd
2 # import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('titanic-dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 plt.hist(data['Age'], bins=50, edgecolor='k')
```

Average time: 0.446 s, Maximum time: 0.446 s, 1 out of 1 shown test case(s) passed

Test case 1: 0.446 ms, Debug, Expected output, Actual output

Terminal, Test cases

36°C Clear

Search

ENG IN

22:56 03-05-2025

[Home](#)
[Learn Anywhere](#)

2023/04/19 09:00
[Support](#)
[Logout](#)

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- Use the Pclass column to group the data for the boxplot.
- Set the title of the plot to "Fare by Pclass".
- Remove the default subtitle with plt.suptitle("").
- Label the x-axis as 'Pclass' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Harris	male	32	1	0	5171	51.0		S
2	1	1	Mr. John Bradley (Florence Briggs Thayer)	female	38	1	0	5199	71.0		C
3	1	3	Mr. Charles (Mrs. Jane) Scott	male	25	0	1	101	17.0		S
4	1	1	Mr. James (Mrs. Mary) Heath	male	35	1	0	1103	53.0		S
5	0	3	Mr. William (Mrs. Mary) Miller	male	30	1	0	104	14.0		S
6	0	3	Mr. James (Mrs. Mary) Miller	male	30	1	0	104	14.0		S
7	0	1	Mr. Timothy J. McCarthy	male	34	0	0	2703	51.0		S
8	0	3	Mr. William (Mrs. Mary) Miller	male	30	1	0	104	14.0		S
9	0	3	Mr. William (Mrs. Mary) Miller	male	30	1	0	104	14.0		S
10	1	1	Mr. James (Mrs. Mary) Heath	male	35	1	0	1103	53.0		S

Sample Data:

```

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, ParCh, Ticket, Fare, Cabin, Embarked
1, 0, 3, Mr. Owen Harris, male, 32, 1, 0, 5171, 51.0, , S
2, 1, 1, Mr. John Bradley (Florence Briggs Thayer), female, 38, 1, 0, 5199, 71.0, , C
3, 1, 3, Mr. Charles (Mrs. Jane) Scott, male, 25, 0, 1, 101, 17.0, , S
4, 1, 1, Mr. James (Mrs. Mary) Heath, male, 35, 1, 0, 1103, 53.0, , S
5, 0, 3, Mr. William (Mrs. Mary) Miller, male, 30, 1, 0, 104, 14.0, , S
6, 0, 3, Mr. James (Mrs. Mary) Miller, male, 30, 1, 0, 104, 14.0, , S
7, 0, 1, Mr. Timothy J. McCarthy, male, 34, 0, 0, 2703, 51.0, , S
8, 0, 3, Mr. William (Mrs. Mary) Miller, male, 30, 1, 0, 104, 14.0, , S
9, 0, 3, Mr. William (Mrs. Mary) Miller, male, 30, 1, 0, 104, 14.0, , S
10, 1, 1, Mr. James (Mrs. Mary) Heath, male, 35, 1, 0, 1103, 53.0, , S

```

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15 data.dropna(inplace=True)

```

Average Time: 0.559 s
Maximum Time: 0.559 s
1 out of 1 shown test case(s) passed

Test case 1 (200 ms) [Debug](#)

Expected output:

Actual output:

[Test cases](#)

20°C

Cool

Search

22°C

22°C

03-05-2025